



Inspiring Excellence

Decentralized Crypto World Bank

A four-tier blockchain banking platform with programmable reserves, hierarchical capital flows, and explainable AI risk support

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DEMONSTRATION FOCUS

Pre-Thesis 2

Architecture · ML benchmarks · Testnet path
Open source prototype

ABSTRACT

Crypto World Bank maps World → National → Local → Client lending onto an EVM ledger with programmable reserves, multi-entity products, and explainable ML risk support.

Keywords: Hierarchical DeFi · Smart contracts · XAI · Group lending

PROBLEM & MOTIVATION

1.4B

UNBANKED

\$4.5T

MSME GAP

2–5d

SETTLE LAG

Flat DeFi ignores institutional hierarchy; development corridors stay opaque and slow. CWB encodes tiered capital and audit trails on-chain.

OBJECTIVES

- 01 Four-tier EVM lending (Ph. I–IV)
- 02 Deposits · savings · group credit
- 03 ML fraud analytics + MCP agent
- 04 Limits · installments · RBAC
- 05 Sepolia / Amoy / Cardona path
- 06 Governance · security docs

METHODOLOGY

Agile MVC (~16 weeks): **Model** = Solidity + PostgreSQL projections; **View** = React/Vite; **Controller** = Express.js + FastAPI. Hardhat tests gate merges (`npm run test:contracts`).

PHASE I

Arch · 3NF schema · Tier 1–3 shells ·
LoanController stub

PHASE II

Loan E2E · SavingsVault ·
GroupLending · interbank pools

PHASE III

Commit–reveal oracle · RF+IF+SHAP
· FX / syndicate

PHASE IV

Gas / MVT metrics · Foundry suite ·
feasibility

On-chain: Solidity 0.8.20 · WorldBankReserve · NationalBank · LocalBank · LoanController · AccessControl roles

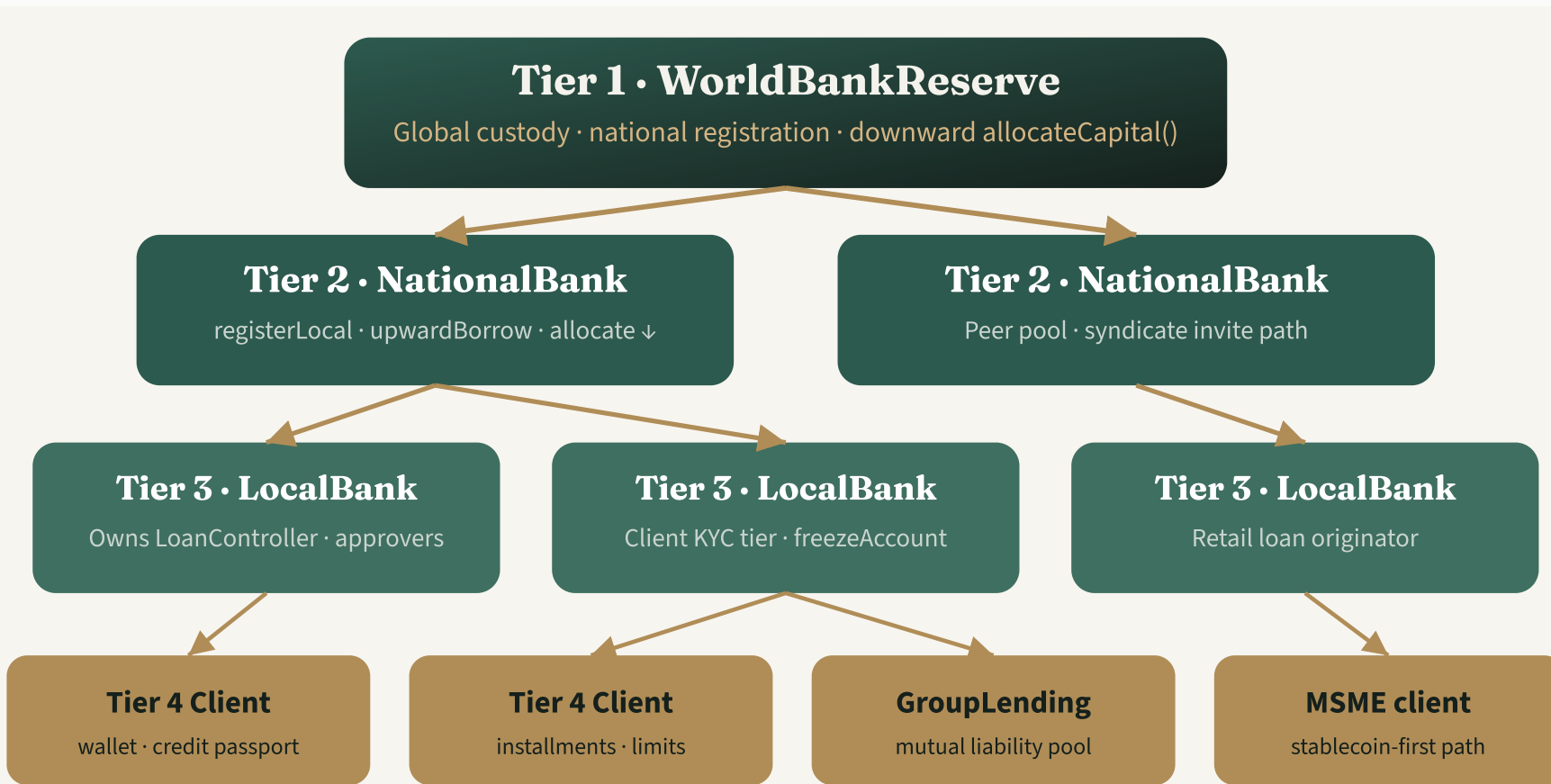
Off-chain: Express API · Prisma/PostgreSQL (51-entity ERD; 16 core MVT) · FastAPI ML · Redis cache

Frontend: React + wallet signing · approver dashboards · optional MCP agent (17 banking tools)

Networks: Ethereum Sepolia (primary) · Polygon Amoy (fallback) · zkEVM Cardona (future)

Oracle: off-chain score → keccak256(s || nonce) commit → reveal on LoanController (human still gates)

SYSTEM SPINE — FOUR TIERS



LoanController state machine (at Local):

Request → Approve → Disburse → Installment schedule → Repay / Overdue / Liquidate (HF<1)

RBAC: WORLD_ADMIN · NATIONAL_ADMIN · LOCAL_APPROVER · CLIENT · optional AGENT session keys

Figure 1. Contract spine — reserve authority cascades; retail credit originates at LocalBank / GroupLendingPool.

CAPITAL MOTION

FOUR ON-CHAIN MOTIONS · MULTI-ENTITY MODULES

Downward

allocateCapital()
Reserve → National
→ Local → Client loan

Reserve-ratio gated
debit lender reserve

LoanController

Peer

InterBankLendingPool
Same-tier liquidity
Local↔Local / Nat↔Nat

Pool utilisation U*
kinked rate @ 80%

Phase II module

Upward

UpwardDepositFacility
Park surplus at parent
yield spread recorded

Repatriation path
vs nostro idle capital

Phase II module

Co-fund

SyndicatedLoan
+ TranchedPool
multi-lender consent

Senior / junior
waterfall recovery

Phase III planned

Also: SavingsVault (reserve-gated withdraw) · GroupLendingPool (solidarity) · stablecoin-first retail path · ≈\$0.011 L2 gas / loan lifecycle

Figure 2. Cross-tier capital motions with named Solidity modules and phase targets.

DEMONSTRATION RESULTS

BCCC-DeFiFraudTrans-2025 subsample (50k txs, ~8.7% fraud, 21 features, wallet-safe split). Scores assist human approvers; disbursement stays gated on-chain.

Split	Prec.	Rec.	F1	AUC
Held-out test	0.874	0.908	0.891	0.991

TN 4,621 · FP 60 · FN 42 · TP 417 (n_{test} = 5,140) · RF + Isolation Forest

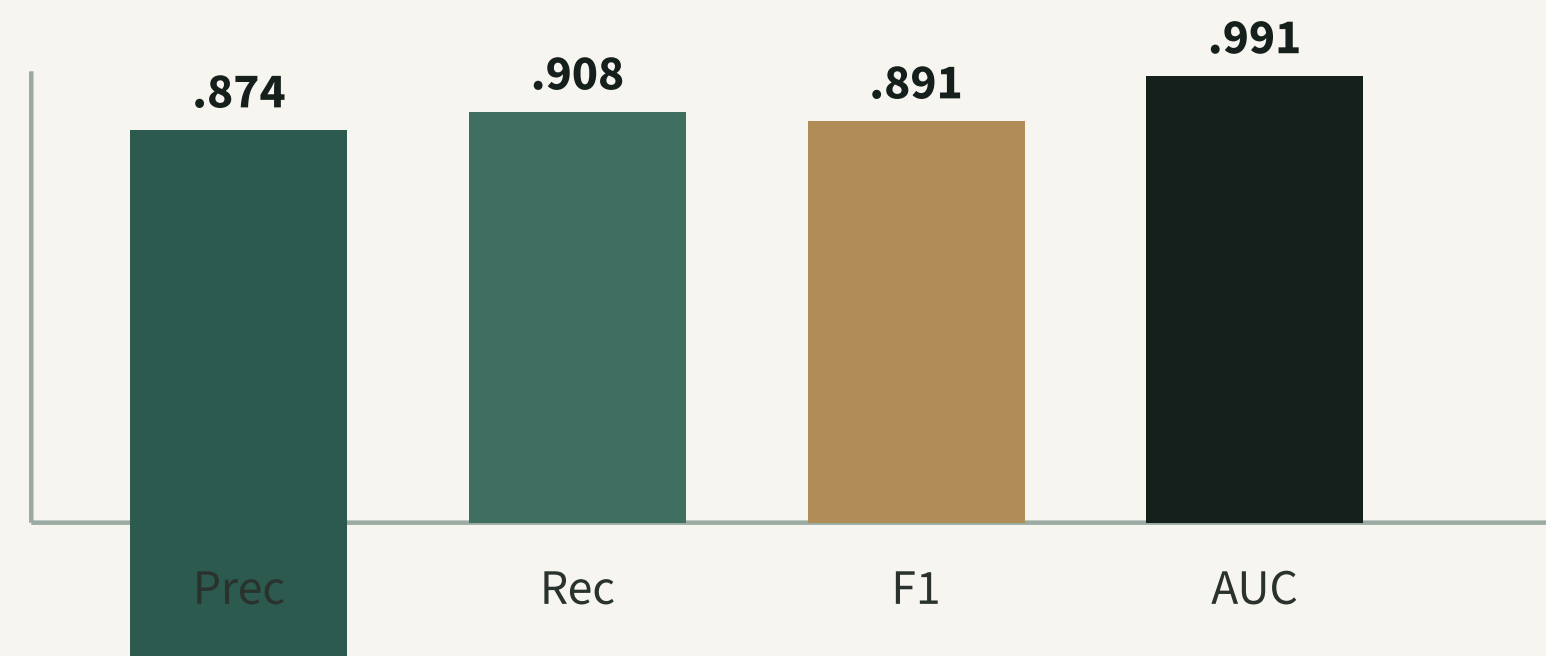


Figure 4. Held-out wallet metrics for the demonstration fraud pipeline.

KEY DESIGN CHOICES

- Stablecoin-first retail path (avoid ETH liability volatility)
- L2 / testnet gas target ≈ \$0.011 per loan lifecycle
- Off-chain RF+IF+SHAP; humans approve; on-chain gates enforce
- Single-chain settlement — multi-chain deferred (bridge risk)
- Optional local MCP agent for demonstration UX

CONCLUSION

Hierarchical development-finance flows can live as auditable smart-contract state, with explainable ML assisting — not replacing — credit judgment. Next: Phase II–IV MVT loan E2E, oracle latency, and CWB-specific retrain.

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